

# 1 Basis Functions and Convergence

Both papers require approximating a continuation value function at each backward time step using a finite set of basis functions. The choice of these functions is a critical implementation decision that affects both accuracy and numerical stability. The two papers treat this choice differently — one empirically and one theoretically — and together they provide a complete picture of when and why regression-based simulation methods work.

## 1.1 Basis Functions in Longstaff–Schwartz (2001)

In the LSM framework, basis functions are used to approximate the conditional expectation function  $F(\omega; t_k)$  — the expected discounted payoff from continuing to hold the option at exercise date  $t_k$ . The formal justification is that when this conditional expectation belongs to the  $L^2$  space of square-integrable functions, that space is a Hilbert space with a countable orthonormal basis, so any such function can be represented as a linear combination of basis elements.

For the vanilla American put, the paper uses a constant plus the first three weighted Laguerre polynomials:

$$L_0(X) = e^{-X/2}, \tag{1}$$

$$L_1(X) = e^{-X/2}(1 - X), \tag{2}$$

$$L_2(X) = e^{-X/2}\left(1 - 2X + \frac{X^2}{2}\right), \tag{3}$$

where  $X = S_t/K$  is the stock price normalised by the strike. The conditional expectation function is then approximated as

$$F(\omega; t_{K-1}) = \sum_{j=0}^{\infty} a_j L_j(X), \tag{4}$$

with coefficients  $a_j$  estimated by ordinary least squares on the in-the-money paths at each exercise date.

For the American–Bermuda–Asian (ABA) option with two state variables  $(S_t, A_t)$ , the basis expands to eight functions: the first two Laguerre polynomials in each of  $S_t/K$  and  $A_t/K$ , together with their cross products up to third order. For the five-asset maximum option in Section 8 of the paper, nineteen basis functions are used: a constant, the first five Hermite polynomials in the maximum asset value, the values and squares of the second through fifth

ranked assets, all pairwise products of adjacent ranked assets, and the product of all five assets.

A central empirical finding in Longstaff and Schwartz (2001) is that basis function choice has surprisingly little effect on pricing accuracy. Section 8.3 of the paper reports that replacing weighted Laguerre polynomials with simple powers  $S$ ,  $S^2$ ,  $S^3$ , Hermite polynomials, or trigonometric functions yields virtually identical results for the vanilla put. The explanation lies in the nature of the regression: since LSM focuses on the *fitted value* rather than individual coefficients, multicollinearity among basis functions does not bias prices, even if it makes individual coefficient estimates unstable. The paper does caution that a nearly singular cross-moment matrix can cause numerical problems, and recommends normalising inputs — for example dividing all prices by the strike  $K$  — to avoid scaling issues.

## 1.2 Basis Functions in Tsitsiklis–Van Roy (2001)

In the TVR framework, basis functions serve a conceptually similar but theoretically more precisely characterised role. Rather than approximating a conditional expectation function directly, they parameterise the  $Q$ -function (continuation value) as a linear combination:

$$\tilde{Q}(x, r) = \sum_{k=1}^K r(k) \phi_k(x) = (\Phi r)(x), \quad (5)$$

where  $\phi_1, \dots, \phi_K$  are fixed basis functions and  $r \in \mathbb{R}^K$  is the parameter vector estimated by regression at each backward step.

The theoretical framework is deliberately agnostic about *which* basis functions are used. What matters is the quantity

$$\epsilon_n^* = \min_r \|\Phi r - Q_n\|_{\pi_n} = \|\Pi_n Q_n - Q_n\|_{\pi_n}, \quad (6)$$

the best approximation error achievable by the chosen basis at time step  $n$ , measured in the norm induced by the trajectory distribution  $\pi_n$ . The algorithm’s total approximation error is bounded in terms of  $\epsilon_n^*$  (see Section 1.3 below), so any basis that makes  $\epsilon_n^*$  small will produce a well-behaved algorithm. The LSM paper’s empirical finding that many different bases give similar results is consistent with this: in one dimension, low-order polynomial and orthogonal polynomial bases all achieve small  $\epsilon^*$  for the smooth continuation value functions that arise in standard option pricing problems.

The practical implication of TVR’s framework becomes most important in higher dimen-

sions. In the ABA option, our LSM implementation uses the eight-function Laguerre basis from the paper, while our TVR-style comparator uses a simpler six-term polynomial basis  $[1, s, s^2, a, a^2, sa]$  where  $s = S_t/K$  and  $a = A_t/K$ . In TVR's framework, the polynomial basis has a larger  $\epsilon^*$  than the Laguerre basis — it approximates the true continuation function less accurately over the relevant state space. This larger  $\epsilon^*$  feeds directly into the error recursion, causing approximation errors to accumulate as the backward induction proceeds, and producing the systematic overpricing observed in the TVR-style comparator results.

### 1.3 Convergence

The two papers offer fundamentally different types of convergence results, reflecting their different goals. LSM provides practical convergence guarantees suited to implementation, while TVR provides theoretical error bounds that explain *why* trajectory-based regression methods work.

#### 1.3.1 Convergence in Longstaff–Schwartz (2001)

LSM convergence is established through two propositions. The first addresses the bias of the algorithm and holds even when the option is continuously exercisable.

**Proposition 1** (LSM downward bias). *For any finite choice of  $M$  basis functions,  $K$  exercise dates, and coefficient vector  $\theta \in \mathbb{R}^{M \times (K-1)}$ , the following inequality holds almost surely:*

$$V(X) \geq \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \text{LSM}(\omega_i; M, K). \quad (7)$$

The intuition is straightforward: the LSM algorithm produces a specific stopping rule, and the true option value  $V(X)$  is the supremum over *all* stopping rules. Any particular stopping rule — including the LSM one — therefore gives a value no greater than the true optimum. This result is important practically: LSM prices are conservative lower bounds on the true American option value, never overstating it in the limit.

**Proposition 2** (LSM convergence). *Assume the option depends on a single state variable  $X$  with support  $(0, \infty)$  following a Markov process, and can be exercised at two dates  $t_1 < t_2$ . If the conditional expectation function  $F(\omega; t_1)$  satisfies*

$$\int_0^\infty e^{-X} F^2(\omega; t_1) dX < \infty \quad \text{and} \quad \int_0^\infty e^{-X} F_X^2(\omega; t_1) dX < \infty,$$

then for any  $\varepsilon > 0$  there exists  $M < \infty$  such that

$$\lim_{N \rightarrow \infty} \Pr \left( \left| V(X) - \frac{1}{N} \sum_{i=1}^N \text{LSM}(\omega_i; M, K) \right| > \varepsilon \right) = 0. \quad (8)$$

This result means that by choosing  $M$  sufficiently large and letting  $N \rightarrow \infty$ , the LSM algorithm converges to any desired degree of accuracy. Longstaff and Schwartz (2001) also validate convergence empirically through an out-of-sample diagnostic test: regression coefficients estimated on one set of paths are applied to a fresh set of paths with different random seeds. The in-sample and out-of-sample values are virtually identical across five seeds, with fewer than 5% of differences exceeding two standard errors.

### 1.3.2 Convergence in Tsitsiklis–Van Roy (2001)

TVR’s convergence analysis distinguishes between two separate questions: the quality of an idealised algorithm with exact expectations and projections, and the convergence of the practical sample-based algorithm to the idealised one.

**Error blow-up with arbitrary representative states.** A key negative result in Tsitsiklis and Van Roy (2001) is that naive approximate value iteration can produce errors that grow *exponentially* in the problem horizon when representative states are chosen arbitrarily. In their Section IV example, an American put is approximated using a single basis function  $\phi(x) = g(x)$  and a projection measure  $\pi$  concentrated at a single fixed point  $1 - \Delta$ . The resulting parameter sequence satisfies

$$r_n \geq \frac{\alpha}{\Delta} (1 - p)(1 - d) r_{n+1}, \quad (9)$$

so that when  $\Delta < \alpha(1 - p)(1 - d)$ , the parameters grow exponentially as  $n$  decreases from  $N$  to 0. Since the true option value is bounded above by  $\alpha^N$ , this exponential growth implies the approximation error diverges.

**Bounded error under trajectory distributions.** The remedy is to sample representative states from simulated trajectories of the underlying process under the risk-neutral measure. Let  $\pi_n$  denote the true risk-neutral distribution of  $x_n$ . By Jensen’s inequality and the tower property of conditional expectations:

$$\|PJ\|_{\pi_n}^2 \leq E[J^2(x_{n+1})] = \|J\|_{\pi_{n+1}}^2, \quad (10)$$

so  $\|PJ\|_{\pi_n} \leq \|J\|_{\pi_{n+1}}$ . This ensures that neither  $P$  nor the projection operator  $\Pi_n$  amplifies errors between time steps. The main convergence result follows.

**Theorem 1** (TVR error bound). *Let  $\pi_0, \dots, \pi_{N-1}$  be the risk-neutral trajectory distributions of  $x_0, \dots, x_{N-1}$ . Define  $\tilde{e}_n = \|\tilde{Q}(\cdot, r_n) - Q_n\|_{\pi_n}$  as the approximation error of the idealised algorithm, and  $\epsilon_n^* = \|\Pi_n Q_n - Q_n\|_{\pi_n}$  as the best achievable error under the chosen basis. Then:*

$$\tilde{e}_{N-1} = \epsilon_{N-1}^*, \quad (11)$$

$$\tilde{e}_n^2 \leq \alpha^2 \tilde{e}_{n+1}^2 + (\epsilon_n^*)^2, \quad n = 0, 1, \dots, N-2. \quad (12)$$

If  $\epsilon_n^* \leq \epsilon$  uniformly, unrolling the recursion gives:

$$\tilde{e}_n \leq \frac{\epsilon}{\sqrt{2(1-\alpha^2)}}, \quad (13)$$

which is *independent of the time horizon  $N$* . This is the central theoretical achievement of Tsitsiklis and Van Roy (2001): error does not grow with the number of time steps. Even the looser bound  $\tilde{e}_n \leq \sqrt{N}\epsilon$  grows only polynomially, not exponentially.

**Theorem 2** (Sample-based convergence). *As the number of simulated trajectories  $m \rightarrow \infty$ , the sample-based parameter vectors  $\hat{r}_n(m)$  converge almost surely to the parameter vectors  $r_n$  of the idealised algorithm:*

$$\hat{r}_n(m) \xrightarrow{a.s.} r_n \quad \text{as } m \rightarrow \infty. \quad (14)$$

The proof applies the strong law of large numbers to the least squares objective: sample averages converge almost surely to population counterparts, so sample regressions converge to population regressions.

## 1.4 Connecting the Two Papers

Proposition 1 establishes that LSM prices never overstate the true American option value in the limit. Theorem 1 establishes that when trajectory-based sampling is used, approximation error remains bounded regardless of horizon. Together these results imply that trajectory-based regression methods produce prices bounded above by the true option value, with the gap depending only on basis quality and not on the number of time steps.

The empirical results of the present study are consistent with both frameworks. Our LSM

prices sit close to but below the binomial American benchmark across all strikes, consistent with the downward bias of Proposition 1. Our TVR-style comparator systematically overshoots the benchmark, consistent with what Theorem 1 predicts when  $\epsilon^*$  is larger: the polynomial basis approximates the continuation function less accurately than the Laguerre basis, and the all-path regression without an ITM filter produces a less focused approximation in the region where the exercise decision matters most. Approximation errors therefore accumulate more aggressively through the backward recursion, producing the systematic overpricing observed across both the vanilla put and the ABA option experiments.